Lesson plan · for the teacher

Nearest neighbours

Data science · Level 3 · Lesson 3

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Tell labels from numbers.
- Classify by nearest neighbours.
- Choose k sensibly.

Activity: The mystery fruit

1. Warm-up (5 min): Describe a new pupil's reading speed and accuracy and ask pupils which group they would put her in, and why.
2. Teach (8 min): Plot a few labelled dots, place a new one and circle its three nearest neighbours.
3. Measure (15 min): Groups weigh fruit, score colour and add their dots to the board.
4. Classify (7 min): Groups classify mystery fruits with k = 1 and k = 3 and compare.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- When did k change the answer?
- Why rescale grams?
- Who should make the final choice for a pupil?

Exit ticket

- Neighbours: blue, red, blue, green, blue. With k = 5, which label wins?

Try it at school tomorrow
Weigh some fruit, score their colour from 1 to 5 and plot them on the board. Bring a mystery fruit and let pupils find its three nearest neighbours.

Practice questions · for you

Nearest neighbours

Data science · Level 3 · Lesson 3

1 **A model chooses a reading group for a new pupil. Does it predict a number or a label?**

◯ A label ◯ Neither ◯ A number

2 **Which of these tasks predict a number, so they are regression? Choose all that apply.**

☐ Predict a reading mark ☐ Choose a reading group ☐ Predict minutes of homework

☐ Decide whether a pupil passes or fails

3 **Find the pupil closest to the new pupil (speed 5, accuracy 7) and print that pupil's group.**

```
pupils = [(2, 3, "red"), (6, 8, "blue"), (9, 2, "green"), (4, 5, "red")]
best = pupils[0]
print(best[2])
```

Write your program:

4 **Why is k = 1 risky?**

◯ It is too slow ◯ One odd example can decide alone ◯ It cannot give a label

5 **If one measure is in grams and the other is 1 to 5, the grams can decide almost everything.**

◯ True ◯ False

✎ **Where do you already sort pupils by comparing them with similar pupils you have taught? What could go wrong?**

Check your answers

Nearest neighbours

1 **A label**

A reading group is a label, so this is classification.

2 **Predict a reading mark; Predict minutes of homework**

A mark and minutes are numbers, so those are regression; a reading group and pass or fail are labels, so those are classification.

3

```
pupils = [(2, 3, "red"), (6, 8, "blue"), (9, 2, "green"), (4, 5, "red")]
best = pupils[0]
bestd = 1000
for p in pupils:
    d = ((p[0] - 5) ** 2 + (p[1] - 7) ** 2) ** 0.5
    if d < bestd:
        bestd = d
        best = p
print(best[2])
```

The pupil at (6, 8) is only about 1.4 away, much closer than the others, so the new pupil gets “blue”.

4 **One odd example can decide alone**

With one neighbour there is no vote, so a single unusual case decides.

5 **True**

Distances in grams are much bigger, so rescale the measures to similar ranges first.